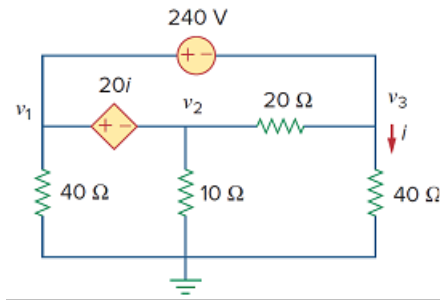


Problem

For the circuit in Fig. 3.69, find v_1 , v_2 , and v_3 using nodal analysis.

Figure 3.69:

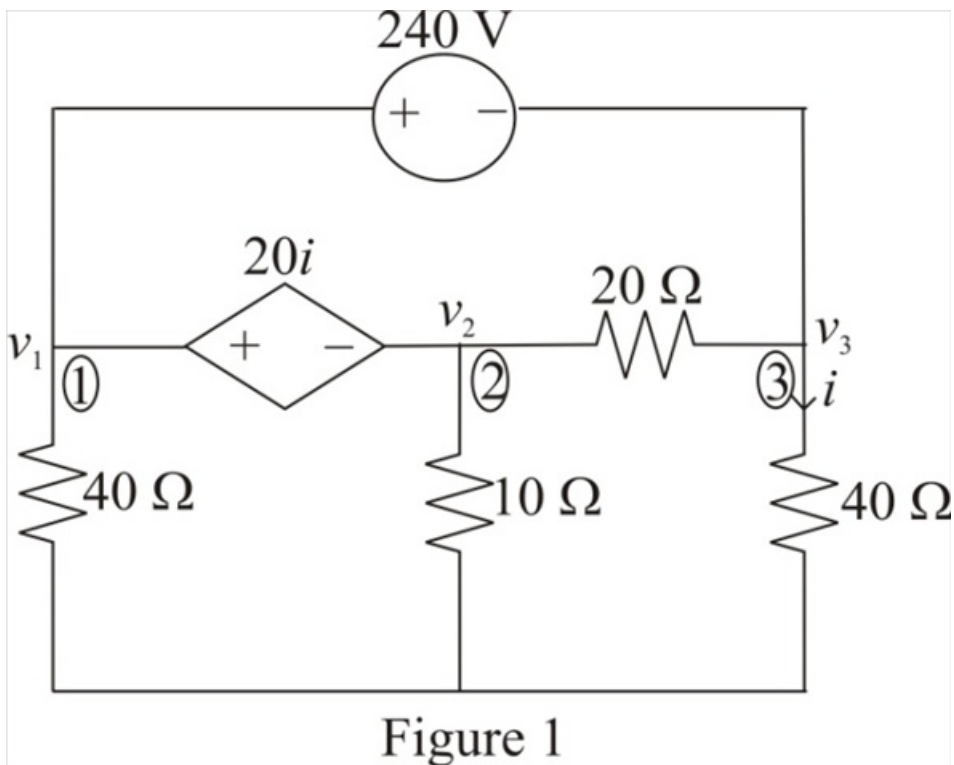


Step-by-step solution

Step 1 of 7

Refer to Figure 3.69 in the text book.

Identify the nodes and voltage in the circuit shown in Figure 1.



Step 2 of 7

Notice that the nodes 1, 2 and 3 represent the super node.

Consider the relation between the voltages v_1 and v_3 .

$$v_1 - v_3 = 240$$

$$v_3 = v_1 - 240 \dots\dots (1)$$

Therefore, the expression for the voltage v_3 is $v_1 - 240$.

Step 3 of 7

Consider the expression for the current i .

$$i = \frac{v_3}{40}$$

Therefore, the expression for the current i is $\frac{v_3}{40}$.

Step 4 of 7

Consider the relation between the voltages v_1 and v_2 .

$$v_1 - v_2 = 20i$$

Substitute $\frac{v_3}{40}$ for i .

$$\begin{aligned} v_1 - v_2 &= 20 \left(\frac{v_3}{40} \right) \\ &= \frac{v_3}{2} \end{aligned}$$

Substitute $v_1 - 240$ for v_3 .

$$v_1 - v_2 = \frac{(v_1 - 240)}{2}$$

$$v_1 - v_2 = \frac{v_1}{2} - 120$$

$$v_2 = 120 - \frac{v_1}{2} + v_1$$

$$v_2 = \frac{v_1}{2} + 120 \dots\dots (2)$$

Therefore, the expression for the voltage v_2 is $\frac{v_1}{2} + 120$.

Step 5 of 7

Apply super node analysis at nodes 1, 2 and 3.

$$\frac{v_1}{40} + \frac{v_2}{10} + \frac{v_2 - v_3}{20} + \frac{v_3 - v_2}{20} + \frac{v_3}{40} = 0$$

$$\frac{v_1 + 4v_2 + 2v_2 - 2v_3 + 2v_3 - 2v_2 + v_3}{40} = 0$$

$$v_1 + 4v_2 + v_3 = 0$$

Substitute $\frac{v_1}{2} + 120$ for v_2 and $v_1 - 240$ for v_3 .

$$v_1 + 4 \left(\frac{v_1}{2} + 120 \right) + (v_1 - 240) = 0$$

$$v_1 + 2v_1 + 480 - 240 + v_1 = 0$$

$$4v_1 + 240 = 0$$

$$4v_1 = -240$$

$$\begin{aligned} v_1 &= \frac{-240}{4} \\ &= -60 \text{ V} \end{aligned}$$

Therefore, the value of the voltage v_1 is, -60 V .

Step 6 of 7

Recall equation (1).

$$v_3 = v_1 - 240$$

Substitute -60 V for v_1 .

$$\begin{aligned} v_3 &= -60 - 240 \\ &= -300 \text{ V} \end{aligned}$$

Therefore, the value of the voltage v_3 is, $\boxed{-300 \text{ V}}$.

Step 7 of 7

Recall equation (2).

$$v_2 = \frac{v_1}{2} + 120$$

Substitute -60 V for v_1 .

$$\begin{aligned} v_2 &= \frac{-60}{2} + 120 \\ &= -30 + 120 \\ &= 90 \text{ V} \end{aligned}$$

Therefore, the value of the voltage v_2 is, $\boxed{90 \text{ V}}$.